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DYNAMICS AND UNIVERSALITY OF AN ISOTHERMAL COMBUSTION PROBLEM IN 2D

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In this paper, the Cauchy problem of the system

$$u_{1,t} = \triangle u_1 - u_1 u_2^m, \quad u_{2,t} = d\triangle u_2 + u_1 u_2^m$$

is studied, where $x \in R^2$, $m \geq 1$ and $d > 0$ is the Lewis number. This system models isothermal combustion (see [7]), and auto-catalytic chemical reaction. We show the global existence and regularity of solutions with non-negative initial values having mild decay as $|x| \to \infty$. More importantly, we establish the exact spatio-temporal profiles for such solutions. In particular, we prove that for $m = 1$, the exact large time behavior of solutions is characterized by a universal, non-Gaussian spatio-temporal profile, with anomalous exponents, due to the fact that quadratic nonlinearity is critical in 2D. Our approach is a combination of iteration using Renormalization Group method, which has been developed into a very powerful tool in the study of nonlinear PDEs largely by the pioneering works of Bricmont, Kupiainen and Lin [6], Bricmont, Kupiainen and Xin, [7], (see also [9]) and key estimates using the PDE method.

Keywords: Auto-catalytic chemical reactions; critical nonlinearity; anomalous exponent; renormalization group; universal spatial-temporal profiles.

Mathematics Subject Classification 2000: 34C20, 34C25, 92E20

1. Introduction

In this paper, we study the initial value problem of reaction-diffusion system

$$u_{1,t} = \triangle u_1 - u_1 u_2^m, \tag{1.1}$$

$$u_{2,t} = d\triangle u_2 + u_1 u_2^m, \tag{1.2}$$

in R^2, where $m \geq 1$ and $d > 0$ is the Lewis number. We assume

$$u_1(x,0) = a_1(x) \geq 0, \quad u_2(x,0) = a_2(x) \geq 0,$$
$$a_1(x),\ a_2(x) \in L^1(R^2) \cap L^\infty(R^2). \tag{1.3}$$

We are concerned with (i) the global existence and regularity of solutions, and (ii) the large time dynamics.

The system (1.1), (1.2) arises as an isothermal combustion model, see [7], and can also be understood as a pre-mixed, isothermal auto-catalytic chemical reaction of the type

$$A + mB \to (m+1)B,$$

under the usual assumption that the isothermal reaction rate is proportional to $u_1 u_2^m$. Here u_1 is the concentration density of reactant A, u_2 is the concentration density of auto-catalyst B. In particular, it contains the two most important auto-catalytic chemical reactions of

$$A + B \to 2B,$$
$$A + 2B \to 3B,$$

in respect to $m = 1$ and $m = 2$ as special cases.

The importance of this model in relation to thermal-diffusive combustion is well-documented in [7]. In chemical reaction theory and biology, it is well known that auto-catalytic chemical reactions play a very important role in complex chain reactions which fulfil many important functions in a living cell and is the focus of intensive research in cell biology.

The study of the system in a bounded domain is carried out by many authors, in particular, see Alikakos [1], Hollis, Martin and Pierre [10], Martin and Pierre [13] and Masuda [14]. Among other things, the authors established boundedness, global existence and large time behavior of solutions. For homogeneous Dirichlet or Neumann boundary conditions, the large time behavior is that (u_1, u_2) converges to a constant vector (c_1, c_2) such that $c_1 \cdot c_2 = 0$, see Masuda [14].

However, our approach is more in line to that of Bricmont, Kupiainen, and Xin [7] and Berlyand and Xin [3], where the Cauchy problem in 1D is studied (see also [18]). As a matter of fact, our main motivation is to extend the pioneering work of Bricmont, Kupiainen, and Xin [7] to 2D and to a wider class of initial values by combining the Renormalization Group (RG) method, and key estimates in using the PDE method.

The existence of traveling front solution in 1D is established by Billingham and Needham in [4, 5] for $m = 1$ and $m = 2$. The authors also study the large-time behavior of solutions using formal methods and numerical computation.

The most important difference in study systems such as (1.1), (1.2) rigorously, as against single equations, is that there is no maximum principle for the systems. We overcome this difficulty by using a combination of PDE analysis (including Maximum Principles to each single equation involved) and the Renormalization Group (RG) method. In particular, deriving *a priori* estimates play a crucial role in our study. It allows us to obtain an exact large time dynamics for a wide class of initial values, demonstrating a universal behavior. But, equally important, it also makes the analysis in general, and Renormalization Group iteration in particular, much simpler and more transparent, demonstrating the power of the RG method in

the study of critical behavior. Another significant aspect of this paper is a detailed analysis of the limiting linear eigenvalue problem for the $m = 1$ case.

We consider the system with initial values $a_1, a_2 \in \mathcal{B}$, where $\mathcal{B}$ is the Banach space of continuous function in R^2 with norm

$$\|f\| = \sup_{x \in R^2} |f(x)|(1 + |x|)^q, \quad q > 2. \tag{1.4}$$

Let ϕ be the Gaussian

$$\phi(x) = \frac{1}{4\pi d} \exp(-|x|^2/4d).$$

For $A > 0$, let ψ_A be the normalized ($\int \psi_A(x)\, d\mu(x) = 1$, see below) principal eigenfunction and $\lambda_A > 0$ the principal eigenvalue, of differential operator

$$\mathcal{L}_A = -\triangle - \frac{x}{2} \cdot \nabla - 1 + A\phi_d(x)$$

on $L^2(\mathbf{R^2}, d\mu)$, with $d\mu(x) = e^{|x|^2/4}\, dx$.

Our main result is the following theorem.

Theorem 1.1. *Suppose the initial values $a_1, a_2 \in \mathcal{B}$ and $a_i \geq 0, \not\equiv 0$, $i = 1, 2$. Let $A = \int_{\mathbf{R^2}} a_1(x) + a_2(x)\, dx > 0$ be the total mass, which is conserved in time. Then the system* (1.1), (1.2) *has a unique global classical solution $(u_1(x,t), u_2(x,t)) \in \mathcal{B} \times \mathcal{B}$ for $\forall\, t \geq 0$. Furthermore,*

(i) *if $m = 1$ and*

$$q > q(A) \equiv 2 + 2\lambda_A, \tag{1.5}$$

there is a positive constant B depending continuously on (a_1, a_2) such that

$$\|t^{1+\lambda_A} u_1(\sqrt{t}\,\cdot, t) - B\psi_A(\,\cdot\,)\| \to 0$$
$$\|t u_2(\sqrt{t}\,\cdot, t) - A\phi(\,\cdot\,)\| \to 0$$

as $t \to \infty$;

(ii) *if $m > 1$ and $q > 2$,*

$$\left\| t u_1(\sqrt{t}x, t) - \frac{A_1}{4\pi} e^{-|x|^2/4} \right\| \to 0 \quad \text{as } t \to \infty$$

and

$$\left\| t u_2(\sqrt{t}x, t) - \frac{A_2}{4\pi d} e^{-|x|^2/4d} \right\| \to 0 \quad \text{as } t \to \infty$$

with $A_1,\ A_2 > 0$ and $A_1 + A_2 = A$.

Remark 1.2. Just as the cubic nonlinearity of $m = 2$ is critical in 1D, the same is true for quadratic nonlinearity of $m = 1$ in 2D. This is the reason why we have the extra decay power λ_A and the non-Gaussian profile of ψ_A. In other words, the scaling law which works for $m > 1$ no longer works for $m = 1$, and thus the appearance of the anomalous exponent λ_A.

Remark 1.3. A distinct feature of our result for $m = 1$ is that we can quantify how large the decaying rate q is for initial values, in terms of total mass A, to qualify for the universal non-Gaussian profile of ψ_A. As a matter of fact, we think our result is optimal, see the Remark 5.9 for further details.

To understand heuristically the result of Theorem 1.1, in particular the critical case of $m = 1$, let us suppose that the nonlinear term in u_1 equation causes some extra decay in time on u_1 in the order of $t^{-\delta}$, where $\delta > 0$, on top of the pure diffusion decay of t^{-1}. This, in turn, results in u_2 having the pure diffusion behavior, since the nonlinear term $u_1 u_2$ is in the order of $t^{-(1+\delta)} u_2$. In the RG terminology, the nonlinear term is irrelevant in the u_2 equation. Then, it is a relatively simple matter to show that

$$tu_2(\sqrt{t}x, t) \to A\phi(x) \quad \text{as } t \to \infty.$$

That is,

$$w_2(y, s) = tu_2(\sqrt{t}x, t), \quad s = \log t \ \text{ and } \ y = \frac{x}{\sqrt{t}},$$

converges to a steady solution of $u_s = -\mathcal{L}u$, where

$$\mathcal{L} = -d\triangle - \frac{x}{2} \cdot \nabla - 1.$$

If we substitute the limiting profile of u_2 into u_1 equation, we have

$$u_{1,t} = \Delta u_1 - t^{-1} A\phi\left(\frac{x}{\sqrt{t}}\right) u_1,$$

which, after a self-similar change of variables

$$u_1(y, s) = tu_1(x, t), \quad y = \frac{x}{\sqrt{t}}, \quad s = \log t,$$

turns into

$$u_{1,s} = -\mathcal{L}_{\mathcal{A}} u_1.$$

Therefore, it is reasonable to guess the large time behavior of u_1 is determined by the first eigenvalue λ_A and the corresponding eigenfunction ψ_A of $\mathcal{L}_{\mathcal{A}}$, so that

$$u_1(y, s)e^{\lambda_A s} \to B\psi_A(y), \quad \text{as } s \to \infty$$

with some constant $B > 0$, provided u_1 has moderate, but sufficient decay in x as $|x| \to \infty$.

But, since we are dealing with an initial value problem with strong nonlinear terms, it is a nontrivial matter to prove the result rigorously. This is particularly true if u_1 has just *modest decay* (not in $L^2(R^2, d\mu)$) as $|x| \to \infty$, which is the case we are dealing with in this paper.

The organization of the paper is as follows. In Sec. 2, we show *a priori* estimates on the solution of the system and consequently establish the global existence. In Sec. 3, we derive the large time dynamics for $m > 1$. In Sec. 4, we consider the

limiting linear eigenvalue problem in relation to the $m = 1$ case and provide some detailed analysis. In Sec. 5, we establish refined bounds for $m = 1$ case and use RG method to prove the convergence of u_1 to a first eigenfunction of $\mathcal{L}_{\mathcal{A}}$, completing the proof of Theorem 1.1.

Throughout the paper, $\|\cdot\|_2$ stands for the $L^2(R^2, d\mu)$-norm of a function, unless otherwise stated, $\|\cdot\|$ the norm defined in (1.4), and other L^p-norm is the standard one. Also, for simplicity of notation, we shall not distinguish generic constant C from line to line.

2. Preliminary Estimates and Global Existence

In this section, we show that u_1 and u_2 have a uniform (in time) $L^p(\mathbf{R^2})$-norm, $1 \le p \le \infty$. Then, it follows from the classical theory that the solutions (u_1, u_2) are smooth and exist globally in time. In this section, all L^p-norms are the standard ones.

First, we collect some simple facts.

Lemma 2.1. *The solutions* (u_1, u_2) *with* $(a_1, a_2) \in (L^1(R^2) \times L^\infty(R^2))^2$ *satisfy the following estimates*:

(i) $\int_{\mathbf{R^2}} (u_1 + u_2)(x,t)\, dx = \int_{\mathbf{R^2}} (a_1 + a_2)(x)\, dx, \quad \forall\, t \ge 0;$

(ii) $\int_{\mathbf{R^2}} u_1(x,t)\, dx \le \int_{\mathbf{R^2}} a_1(x)\, dx, \quad \int_{\mathbf{R^2}} u_2(x,t)\, dx \ge \int_{\mathbf{R^2}} a_2(x)\, dx \quad \forall\, t \ge 0;$

(iii) $\int_0^\infty \int_{\mathbf{R^2}} u_1 u_2^m(x,t)\, dxdt < +\infty.$

Furthermore,

(iv) $0 \le u_1(x,t) \le \|a_1\|_\infty,\ u_1(x,t) \le \overline{u}_1(x,t)$ *and* $u_2(x,t) \ge \underline{u}_2(x,t),\ \forall\, t \ge 0,$ *where* $\underline{u}_2$ *is a solution of the heat equation*

$$\underline{u}_{2,t} = d\triangle \underline{u}_2, \quad \underline{u}_2|_{t=0} = a_2(x),$$

and $\overline{u}_1$ *is a solution of*

$$\overline{u}_{1,t} = \triangle \overline{u}_1 - \overline{u}_1 \underline{u}_2^m, \quad \overline{u}_1|_{t=0} = a_1(x).$$

Proof. Simple integration yields (i)–(iii). (iv) is obtained by direct use of maximum principle.

The key estimates of this section are stated in the following lemma.

Lemma 2.2. *The solutions* (u_1, u_2) *of* (1.1)–(1.2) *are uniformly bounded in time in* $L^p(\mathbf{R^2})$ *norm*:

$$\|u_1(\cdot\,,\,t)\|_{L^p} + \|u_2(\cdot\,,\,t)\|_{L^p} \le C(a_1, a_2, p) < \infty, \quad 1 \le p < \infty, \tag{2.1}$$

where $C(a_1, a_2, p)$ *is a constant depending on initial data and* p *is a positive integer.*

Proof. It is clear by Lemma 2.1 that the u_1 bound holds. We proceed to show the bound for u_2. By standard local existence theory, (u_1, u_2) are classical solutions local in time and all $L^p(\mathbf{R^2})$-norm of (u_1, u_2) and the $L^2(\mathbf{R^2})$-norm of $(\nabla u_1, \nabla u_2)$ are finite and continuous in time. Therefore, we can freely perform integration by parts. By multiplying (1.2) by pu_2^{p-1} with $p \geq 2$, integrating over $\mathbf{R^2} \times (0, t)$, we get

$$\int_{\mathbf{R^2}} u_2^p \, dx = \int_{\mathbf{R^2}} a_2^p \, dx - d \int_0^t \int_{\mathbf{R^2}} p(p-1)|\nabla u_2|^2 u_2^{p-2} \, dx d\tau + \int_0^t \int_{\mathbf{R^2}} p u_1 u_2^{m+p-1} \, dx d\tau. \tag{2.2}$$

In addition, with the help of integration by parts, we derive the identity

$$\begin{aligned}
&\frac{d}{dt} \int_{\mathbf{R^2}} (u_1^2 + u_1) u_2^p \, dx d\tau \\
&\quad = \int_{\mathbf{R^2}} (1 + 2u_1)(\triangle u_1 - u_1 u_2^m) u_2^p \, dx + \int_{\mathbf{R^2}} (u_1^2 + u_1) p u_2^{p-1} (d \triangle u_2 + u_1 u_2^m) \, dx \\
&\quad = -2 \int_{\mathbf{R^2}} |\nabla u_1|^2 u_2^p \, dx - \int_{\mathbf{R^2}} (1 + 2u_1) p u_2^{p-1} \nabla u_1 \cdot \nabla u_2 \\
&\qquad - \int_{\mathbf{R^2}} (1 + 2u_1) u_1 u_2^{m+p} \, dx - d \int_{\mathbf{R^2}} (1 + 2u_1) p u_2^{p-1} \nabla u_1 \cdot \nabla u_2 \\
&\qquad - d \int_{\mathbf{R^2}} (u_1^2 + u_1) p(p-1) u_2^{p-2} |\nabla u_2|^2 \, dx + \int_{\mathbf{R^2}} (u_1^2 + u_1) p u_2^{p+m-1} u_1 \, dx \\
&\quad = I + II + III + IV + V + VI.
\end{aligned} \tag{2.3}$$

It is clear that

$$\begin{aligned}
I + II + IV &\leq (1 + 2\|a_1\|_\infty)(1 + d) \int_{\mathbf{R^2}} |\nabla u_1 \cdot \nabla u_2| p u_2^{p-1} - 2 \int_{\mathbf{R^2}} |\nabla u_1|^2 u_2^p \, dx \\
&\leq \int_{\mathbf{R^2}} |\nabla u_1|^2 u_2^p \, dx + C(\|a_1\|_\infty, p) \int_{\mathbf{R^2}} |\nabla u_2|^2 u_2^{p-2} \, dx \\
&\qquad - 2 \int_{\mathbf{R^2}} |\nabla u_1|^2 u_2^p \, dx \\
&\leq C(\|a_1\|_\infty, p) \int_{\mathbf{R^2}} |\nabla u_2|^2 u_2^{p-2} \, dx - \int_{\mathbf{R^2}} |\nabla u_1|^2 u_2^p \, dx.
\end{aligned}$$

Moreover,

$$III \leq - \int_{\mathbf{R^2}} u_1 u_2^{m+p} \, dx, \quad V \leq 0$$

and

$$VI \leq (\|a_1\|_\infty + \|a_1\|_\infty^2) \int_{\mathbf{R^2}} p u_1 u_2^{m+p-1} \, dx.$$

An integration of (2.3) from 0 to t yields

$$\begin{aligned}\left(\int_{\mathbf{R^2}}(u_1^2+u_1)u_2^p\,dx\right)(t) &\le \int_{\mathbf{R^2}}(a_1^2+a_1)a_2^p\,dx + C(p,a_1)\int_0^t\int_{\mathbf{R^2}}|\nabla u_2|^2u_2^{p-2}\,dxd\tau \\ &\quad + (\|a_1\|_\infty + \|a_1\|_\infty^2)\int_0^t\int_{\mathbf{R^2}} pu_1u_2^{m+p-1}\,dxd\tau \\ &\quad - \int_0^t\int_{\mathbf{R^2}} u_1u_2^{m+p}\,dxd\tau - \int_0^t\int_{\mathbf{R}}|\nabla u_1|^2u_2^p\,dxd\tau. \qquad (2.4)\end{aligned}$$

The combination of (2.2) and (2.4) then gives

$$\begin{aligned}&\int_{\mathbf{R^2}}u_2^p + \int_0^t\int_{\mathbf{R^2}}(|\nabla u_2|^2u_2^{p-2} + |\nabla u_1|^2u_2^p + u_1u_2^{m+p})\,dxd\tau \\ &\quad \le C(a_1,a_2,p)\left(1+\int_0^t\int_{\mathbf{R^2}} u_1u_2^{m+p-1}\,dxd\tau\right). \qquad (2.5)\end{aligned}$$

For $p=1$, we want to show

$$\int_0^t\int_{\mathbf{R^2}}(u_1u_2^{m+1} + |\nabla u_1|^2u_2)\,dxd\tau \le C(a_1,a_2)\left(1+\int_0^t\int_{\mathbf{R^2}} u_1u_2^m\,dxd\tau\right). \qquad (2.6)$$

Proceed as the case of $p\ge 2$, we have

$$\begin{aligned}&\frac{d}{dt}\int_{\mathbf{R^2}}(u_1^2+u_1)u_2\,dx \\ &\quad = -2\int_{\mathbf{R^2}}|\nabla u_1|^2u_2\,dx - \int_{\mathbf{R^2}}(1+2u_1)\nabla u_1\cdot\nabla u_2 - \int_{\mathbf{R^2}}(1+2u_1)u_1u_2^{m+1}\,dx \\ &\qquad - d\int_{\mathbf{R}}(1+2u_1)\nabla u_1\cdot\nabla u_2\,dx + \int_{\mathbf{R^2}}(u_1^2+u_1)u_1u_2^m\,dx \\ &\quad \le -2\int_{\mathbf{R^2}}|\nabla u_1|^2u_2\,dx + C(a_1,a_2)\int_{\mathbf{R^2}}(|\nabla u_1\cdot\nabla u_2| + u_1u_2^m) \\ &\qquad - \int_{\mathbf{R^2}} u_1u_2^{m+1}\,dx \\ &\quad \le -2\int_{\mathbf{R^2}}|\nabla u_1|^2u_2\,dx + C(a_1,a_2)\int_{\mathbf{R^2}}(\epsilon^{-1}|\nabla u_1|^2 + \epsilon|\nabla u_2|^2 + u_1u_2^m) \\ &\qquad - \int_{\mathbf{R^2}} u_1u_2^{m+1}\,dx \qquad (2.7)\end{aligned}$$

for any $\epsilon>0$. Now, integrating the above inequality from 0 to t gives

$$\begin{aligned}&\int_{\mathbf{R^2}}(u_1^2+u_1)u_2\,dx + \int_0^t\int_{\mathbf{R^2}}(2|\nabla u_1|^2u_2 + u_1u_2^{m+1})\,dxd\tau \\ &\quad \le C(a_1,a_2)\left(1+\int_0^t\int_{\mathbf{R^2}}(\epsilon^{-1}|\nabla u_1|^2 + \epsilon|\nabla u_2|^2 + u_1u_2^m)\right)dxd\tau. \qquad (2.8)\end{aligned}$$

With the help of (2.2) when $p = 2$, we obtain

$$2d \int_0^t \int_{\mathbf{R^2}} |\nabla u_2|^2 \, dx d\tau \leq \int_{\mathbf{R^2}} a_2^2 \, dx + 2 \int_0^t \int_{\mathbf{R}} u_1 u_2^{m+1} \, dx d\tau. \tag{2.9}$$

Similarly, by multiplying (1.1) by $2u_1$ and integrating over $\mathbf{R^2} \times (0, t)$, we have

$$\int_{\mathbf{R^2}} u_1^2 \, dx = \int_{\mathbf{R^2}} a_1^2 \, dx - 2 \int_0^t \int_{\mathbf{R^2}} |\nabla u_1|^2 \, dx d\tau - 2 \int_0^t \int_{\mathbf{R^2}} u_1^2 u_2^m \, dx d\tau, \tag{2.10}$$

which implies that

$$2 \int_0^t \int_{\mathbf{R^2}} |\nabla u_1|^2 \, dx d\tau \leq \int_{\mathbf{R^2}} a_1^2 \, dx. \tag{2.11}$$

Combining (2.8), (2.9) and (2.11) with sufficiently small ϵ gives the desired inequality (2.6). It is easy to see that the lemma follows from a simple induction on p using (2.5) and (2.6).

Lemma 2.3. *The following estimates for the derivative of* (u_1, u_2) *hold if* $t \geq t_0 > 0$:

$$\|\nabla u_1(\cdot\,, t)\|_2 + \|\nabla u_2(\cdot\,, t)\|_2 \leq C(a_1, a_2) < \infty, \tag{2.12}$$

where $t_0 > 0$ *is arbitrary.*

Proof. The above estimates follow by standard techniques once the L^p estimate of (u_1, u_2) is established. For completeness, we include the details here. First, we have from (2.10),

$$\int_{\mathbf{R^2}} u_1^2 \, dx + \int_0^t \int_{\mathbf{R^2}} |\nabla u_1|^2 \, dx d\tau \leq \int_{\mathbf{R^2}} a_1^2 \, dx.$$

Similarly, we have

$$\int_{\mathbf{R^2}} u_2^2 \, dx + \int_0^t \int_{\mathbf{R^2}} |\nabla u_2|^2 \, dx d\tau \leq \int_0^t \int_{\mathbf{R^2}} u_1 u_2^{m+1} \, dx d\tau + \int_{\mathbf{R^2}} a_2^2 \, dx \leq C(a_1, a_2)$$

by Lemma 2.2. By Fubini's theorem, there exists $t_1 \in [0, t_0]$ such that

$$\left(\int_{\mathbf{R^2}} |\nabla u_1|^2 + |\nabla u_2|^2 \, dx \right)(t_1) \leq C(a_1, a_2)/t_0. \tag{2.13}$$

Next, multiplying (1.1) by $\triangle u_1$ and integrating over $\mathbf{R^2}$ yield

$$\begin{aligned} -\frac{1}{2} \frac{d}{dt} \|\nabla u_1\|_2^2 &= \int_{\mathbf{R^2}} (\triangle u_1)^2 \, dx - \int_{\mathbf{R^2}} u_1 u_2^m \triangle u_1 \, dx \\ &= \int_{\mathbf{R^2}} (\triangle u_1)^2 \, dx + \int_{\mathbf{R^2}} u_2^m |\nabla u_1|^2 \, dx + m \int_{\mathbf{R^2}} u_1 u_2^{m-1} \nabla u_1 \cdot \nabla u_2 \, dx. \end{aligned} \tag{2.14}$$

By similar means, we have

$$-\frac{1}{2}\frac{d}{dt}\|\nabla u_2\|_2^2$$
$$= d\int_{\mathbf{R^2}}(\triangle u_2)^2\,dx - m\int_{\mathbf{R^2}} u_1 u_2^{m-1}|\nabla u_2|^2\,dx - \int_{\mathbf{R^2}} u_2^m \nabla u_1\cdot\nabla u_2\,dx. \tag{2.15}$$

Adding (2.14) and (2.15), and integrate from t_1 to t then gives

$$(\|\nabla u_1\|_2^2 + \|\nabla u_2\|_2^2)(t)$$
$$\le (\|\nabla u_1\|_2^2 + \|\nabla u_2\|_2^2)(t_1) + \int_{t_1}^{t}\int_{\mathbf{R^2}} u_2^m|\nabla u_1\cdot\nabla u_2|\,dxd\tau$$
$$+ m\int_{t_1}^{t}\int_{\mathbf{R^2}} u_1 u_2^{m-1}|\nabla u_2|^2\,dxd\tau + m\int_{t_1}^{t}\int_{\mathbf{R^2}} u_1 u_2^{m-1}|\nabla u_1\cdot\nabla u_2|\,dxd\tau. \tag{2.16}$$

We now derive bounds for each of the last three terms on the right-hand side of (2.16). Let M be the smallest positive integer bigger or equal to m.

$$\int_{t_1}^{t}\int_{\mathbf{R^2}} u_2^m|\nabla u_1\cdot\nabla u_2|\,dxd\tau$$
$$\le \left(\int_{t_1}^{t}\int_{\mathbf{R^2}} u_2^m|\nabla u_1|^2\right)^{1/2}\left(\int_{t_1}^{t}\int_{\mathbf{R^2}} u_2^m|\nabla u_2|^2\right)^{1/2}$$
$$\le \left(\int_{t_1}^{t}\int_{\mathbf{R^2}} u_2^M|\nabla u_1|^2\right)^{m/2M}\left(\int_{t_1}^{t}\int_{\mathbf{R^2}} |\nabla u_1|^2\right)^{(M-m)/2M}$$
$$\times\left(\int_{t_1}^{t}\int_{\mathbf{R^2}} u_2^M|\nabla u_2|^2\right)^{m/2M}\left(\int_{t_1}^{t}\int_{\mathbf{R^2}} |\nabla u_2|^2\right)^{(M-m)/2M} \tag{2.17}$$

by Hölder's inequality. Similarly,

$$\int_{t_1}^{t}\int_{\mathbf{R^2}} u_1 u_2^{m-1}|\nabla u_2|^2\,dxd\tau$$
$$\le \|a_1\|_\infty\int_{\mathbf{R^2}} u_2^{m-1}|\nabla u_2|^2\,dxd\tau$$
$$\le \|a_1\|_\infty\left(\int_{t_1}^{t}\int_{\mathbf{R^2}} u_2^{M-1}|\nabla u_2|^2\,dxd\tau\right)^{\sigma}\left(\int_{t_1}^{t}\int_{\mathbf{R^2}} |\nabla u_2|^2\,dxd\tau\right)^{1-\sigma}, \tag{2.18}$$

where $\sigma = (m-1)/(M-1)$ and

$$\begin{aligned}
&\int_{t_1}^{t}\int_{\mathbf{R^2}} u_1 u_2^{m-1}|\nabla u_1 \cdot \nabla u_2|\, dxd\tau \\
&\quad \le \|a_1\|_\infty \int_{t_1}^{t}\int_{\mathbf{R^2}} u_2^{m-1}|\nabla u_1 \cdot \nabla u_2|\, dxd\tau \\
&\quad \le \|a_1\|_\infty \left(\int_{t_1}^{t}\int_{\mathbf{R^2}} u_2^{M-1}|\nabla u_1|^2\, dxd\tau\right)^{\sigma/2} \left(\int_{t_1}^{t}\int_{\mathbf{R^2}} |\nabla u_1|^2\, dxd\tau\right)^{(1-\sigma)/2} \\
&\qquad \times \left(\int_{t_1}^{t}\int_{\mathbf{R^2}} u_2^{M-1}|\nabla u_2|^2\, dxd\tau\right)^{\sigma/2} \left(\int_{t_1}^{t}\int_{\mathbf{R^2}} |\nabla u_2|^2\, dxd\tau\right)^{(1-\sigma)/2}. \qquad (2.19)
\end{aligned}$$

With the help of (2.16)–(2.19), and (2.5) and (2.6) in Lemma 2.2, we obtain

$$(\|\nabla u_1\|_2^2 + \|\nabla u_2\|_2^2)(t) \le (\|\nabla u_1\|_2^2 + \|\nabla u_2\|_2^2)(t_1) + C(a_1, a_2).$$

Hence, by (2.13),

$$(\|\nabla u_1\|_2^2 + \|\nabla u_2\|_2^2)(t) \le C(a_1, a_2)$$

for all $t \ge t_0$.

Proposition 2.4. *The system* (1.1), (1.2) *has a unique classical solution satisfying, for all* $t > 0$,

$$\|u_1\|_{L^p} + \|u_2\|_{L^p} \le C(a_1, a_2), \quad 1 \le p \le \infty, \qquad (2.20)$$

where the constant depends only on the initial data $(a_1, a_2) \in (L^1(\mathbf{R^2}) \cap L^\infty(\mathbf{R^2}))^2$.

Proof. The bound for u_1 follows directly from Lemmas 2.1 and 2.2. For u_2, the classical theory of local existence yields the bound for small t, say $t \le t_0$, where $t_0 > 0$. For $t > t_0$, if we write Eq. (1.2) as

$$u_{2,t} - d\triangle u_2 = f,$$

the standard boot-strapping argument yields

$$\|u_2\|_{2,2} \le C(\|u_2\|_{1,2} + \|f\|_2),$$

where $\|\cdot\|_{2,2}$ is the norm of Sobolev space $W^{2,2}(\mathbf{R^2})$ and $\|\cdot\|_{1,2}$ that of $W^{1,2}(\mathbf{R^2})$, with $C = C(n, d)$. The Sobolev embedding, (2.5) for $p = 2$ and Lemma 2.3 give us

$$\|u_2\|_\infty \le C(a_1, a_2).$$

This completes the proof of the proposition.

3. The Case of $m > 1$

In this section, we show that the solutions (u_1, u_2), when $m > 1$, have the large-time dynamics as solutions of pure heat equations.

First, it is clear by maximum principle that

$$\|u_1\|_\infty \le C(\|a_1\|)(1+t)^{-1}, \quad \forall\, t \ge 0. \tag{3.1}$$

Second,

$$\begin{aligned} u_2(x,t) &= \int_{\mathbf{R^2}} H(y,t)a_2(x-y)\,dy + \int_0^t\int_{\mathbf{R^2}} H(y,s)u_1u_2^m(x-y,t-s)\,dyds \\ &\le \frac{C(\|a_2\|)}{1+t} + \left(\int_0^1\int_{\mathbf{R^2}} + \int_1^t\int_{\mathbf{R^2}}\right) H(y,s)u_1u_2^m(x-y,t-s)\,dyds, \end{aligned}$$

where

$$H(x,t) = \frac{1}{4\pi dt}e^{-x^2/4dt}$$

is the heat kernel.

$$\begin{aligned} &\int_0^1\int_{\mathbf{R^2}} H(y,s)u_1u_2^m(x-y,t-s)\,dyds \\ &\quad\le \int_0^1 \|u_2\|_\infty^m\|u_1\|_\infty(t-s)\,ds \\ &\quad\le C(\|a_1\|,\|a_2\|)\int_0^1 \frac{1}{1+t-s}\,ds \\ &\quad\le \frac{C(\|a_1\|,\|a_2\|)}{t} \end{aligned}$$

using (3.1) and the L^∞ bound $\|u_2\|_\infty \le C(\|a_1\|,\|a_2\|)$.

$$\begin{aligned} &\int_1^t\int_{\mathbf{R^2}} H(y,s)u_1u_2^m(x-y,t-s)\,dyds \\ &\quad\le \int_1^t s^{-1}\|u_1u_2^{m-1}\|_\infty(t-s)ds\int_{\mathbf{R^2}} u_2\,dy \\ &\quad\le C(\|a_1\|,\|a_2\|)\int_1^t s^{-1}\cdot\frac{1}{1+t-s}ds \\ &\quad\le C(\|a_1\|,\|a_2\|)2\log t/(1+t). \end{aligned} \tag{3.2}$$

Therefore,

$$\|u_2\|_\infty \le C(\|a_1\|,\|a_2\|)\log t/(1+t).$$

Substitute this inequality back into (3.2), we have

$$\begin{aligned}
&\int_1^t \int_{\mathbf{R^2}} H(y,s) u_1 u_2^m (x-y, t-s)\, dyds \\
&\quad \le C(\|a_1\|, \|a_2\|) \int_1^t s^{-1} \frac{(\log(1+t-s))^{m-1}}{(1+t-s)^m} ds \\
&\quad \le C/(1+t).
\end{aligned}$$

The last inequality can be established, for instance, by breaking the integral into two parts on $[1, t^{2/(1+m)}]$ and $[t^{2/(1+m)}, t]$ and estimating each part. Hence,

$$\|u_2\|_\infty \le C(\|a_1\|, \|a_2\|)/(1+t). \tag{3.3}$$

It is then clear that both u_1 and u_2 satisfy an equation in the form

$$u_t = D\triangle u + O((1+t)^{-m})u,$$

with D a positive constant. By classical theory of heat equation, the nonlinear term is irrelevant and the large-time dynamics for each of them is characterized by a pure heat equation. That is,

$$\left\| t u_1(\sqrt{t}x, t) - \frac{A_1}{4\pi} e^{-|x|^2/4} \right\| \to 0 \quad \text{as } t \to \infty$$

and

$$\left\| t u_2(\sqrt{t}x, t) - \frac{A_2}{4\pi d} e^{-|x|^2/4d} \right\| \to 0 \quad \text{as } t \to \infty$$

with $A_1,\ A_2 > 0$ and $A_1 + A_2 = A$. This is the exact statement of (ii) in Theorem (1.1).

4. The Linear Eigenvalue Problem

In this section, we study the linear eigenvalue problem

$$\mathcal{L}_{\mathcal{A}} w = \lambda_A w,$$

where $\lambda_A > 0$ is the first eigenvalue of $\mathcal{L}_{\mathcal{A}}$. First, we list some known facts from classical functional analysis and other sources, for details see [7, 8].

(i) The spectrum of $\mathcal{L}_{\mathcal{A}}$, as considered in $L^2(R^2, d\mu)$, consists of eigenvalues only and the eigenfunctions form a complete orthogonal set in that space.
(ii) The first eigenvalue $\lambda_A > 0$ is non-degenerate and the corresponding eigenfunction is positive, radial with $w = w(r)$, where $r = |y|$ and $w'(0) = 0$.
(iii) λ_A depends continuously on A, is an increasing function of A and is strictly less than $A/4\pi d$.

The main purpose of this section is to show that the eigenfunction w has the asymptotics, as $y \to \infty$,

$$w(r) = ce^{-r^2/4}|r|^{2\lambda_A} + h.o.t$$

with c a positive constant.

As a matter of fact, we shall consider the more general case of

$$\begin{aligned} w'' + \frac{r}{2}w' + \frac{w'}{r} + w - A\phi w &= -\lambda w, \\ w'(0) = 0, \quad w(r) > 0, \quad \forall\, r \geq 0, \end{aligned} \tag{4.1}$$

and prove that a positive solution with $\lambda > 0$ has the asymptotics as above when λ_A is replaced by λ, provided that $w \in L^2(R^2, d\mu)$.

If we integrate Eq. (4.1), we have

$$rw'(r) + \frac{r^2}{2}w = \int_0^r \nu(A\phi - \lambda)w\, d\nu. \tag{4.2}$$

If there exists $r_0 > 0$ such that

$$w'(r_0) + r_0 w(r_0)/2 < 0 \quad \text{and} \quad A\phi(y) - \lambda < 0, \qquad \forall\, r \geq r_0,$$

then $g(r) = w'(r) + rw/2$ satisfies $g(r_0) \leq 0$ and $rg'(r) + g(r) < 0$, $\forall\, r \geq r_0$. Hence, there exists $\eta > 0$ with the property $rg(r) < -\eta$, $\forall\, r \geq r_0 + 1 \equiv r_1$. An integration then yields

$$e^{r^2/4}w(r) - e^{r_1^2/4}w(r_1) < -\eta \int_{r_1}^r e^{\nu^2/4}\nu^{-1}\, d\nu,$$

which in turn implies

$$w(r) < -\frac{\eta}{2r^2}, \quad \forall\, r \gg 1.$$

A contradiction! Thus, $g(r) > 0$ for all r large. In addition, it is easy to see that w' cannot change sign for r large. This, together with $g(r) > 0$, shows $w'(r) < 0$ for all r large. An inspection of (4.2) gives $g(r) > 0$ for all $r > 0$. In consequence, $g'(r) \to 0$ as $r \to \infty$.

To derive the exact asymptotics, we make a change of variables. Let

$$z = e^{r^2/4}w.$$

We know from the above analysis that $z, z' > 0$ $\forall\, r > 0$. The equation for z is

$$z'' - \frac{r}{2}z' + \frac{1}{r}z' = (A\phi - \lambda)z.$$

If $z \to M > 0$ as $r \to \infty$, we would have, upon an integration,

$$z' > \lambda z/r, \quad \forall\, r \gg 1.$$

Clearly, this is in contradiction with $z \to M > 0$ as $r \to \infty$. Hence, $z \to \infty$ as $r \to \infty$.

Let $G(y) = rz'/2 - \lambda z$,

$$G' = \frac{r}{2}z'' + \left(\frac{1}{2} - \lambda\right)z' = \left(\frac{r}{2} - \frac{2\lambda}{r}\right)G - \frac{2\lambda^2}{r}z + \frac{r}{2}A\phi z.$$

We show that $G > 0$ for $r \gg 1$. Otherwise, $G < 0$ for all r large. Furthermore,

$$Ge^{-r^2/4}r^{2\lambda} < -C < 0, \quad \forall\, r \gg 1.$$

In consequence,

$$z > \frac{C}{\lambda}e^{r^2/4}r^{-2\lambda} \Leftrightarrow w(r) > \frac{C}{\lambda}r^{-2\lambda}, \quad \forall\, r \gg 1,$$

which is in clear contradiction with $w \in L^2(R^2, d\mu)$. Hence, $G > 0$ for all $r \gg 1$.

We now show

$$zr^{-2\lambda} \to C > 0 \quad \text{as } r \to \infty.$$

For this purpose, we make another change of variables by letting

$$\sigma(t) = zr^{-2\lambda}, \quad t = \log r.$$

The equation for σ, using $\sigma' = d\sigma/dt$, is as follows:

$$\sigma'' + \left(\frac{-e^{2t}}{2} + 4\lambda\right)\sigma' + 4\lambda^2\sigma = e^{2t}A\phi\sigma + \frac{e^{2t}}{\lambda}\sigma. \tag{4.3}$$

First, we observe that for any $\delta > 0$, $\sigma' < \delta\sigma$ if $t \gg 1$. Otherwise, we would have, using (4.3),

$$\frac{\sigma''}{\sigma'} > 0, \quad (\sigma' - \delta\sigma)' > 0 \quad \forall\, t \gg 1.$$

Hence,

$$\frac{\sigma''}{\sigma'} > \left(\frac{1}{2} - \epsilon\right)e^{2t} \quad \text{for any } \epsilon > 0 \quad \text{if } t \gg 1,$$

an integration of which yields

$$\log \sigma' > \frac{1}{2}\left(\frac{1}{2} - \epsilon\right)e^{2t}.$$

Moreover, we see that

$$\sigma' - \mu\sigma > 0, \quad (\sigma' - \mu\sigma)' > 0,$$

for any $\mu > 0$, if $t \gg 1$. Consequently, we would have $z' \geq \mu z$ if $y \gg 1$. A contradiction! It is clear from the above argument that we must have $4\lambda^2\sigma > \sigma' e^{2t}/4$ for all $t \gg 1$. An integration of which yields σ is bounded from above. Hence, $\sigma \to C > 0$ as $t \to \infty$.

The asymptotics of w is thus established.

5. The Case of $m = 1$

In this section, we use the Renormalization Group (RG) method, in combination with the *a priori* estimates derived in Sec. 2 and below to derive the spatio-temporal profile of general solutions for the case of $m = 1$, and therefore completing the proof of Theorem 1.1.

First, we summarize some basic facts on u_1 and u_2.

Proposition 5.1. (i) *If* $0 \le a_1, a_2 \in \mathcal{B}$, *then* $u_1(\cdot\,,t), u_2(\cdot\,,t) \in \mathcal{B}$ *for any* $t > 0$, *and there exists* $C = C(q, \|a_1\|, \|a_2\|) > 0$ *such that*

$$u_1(x,t) + u_2(x,t) \le e^{Ct}(1+|x|)^{-q}.$$

(ii) *For any* $t_0 > 0$, *there exists* $\delta > 0$ *such that*

$$u_2(x,t) \ge \delta(1+t)^{-1}\phi\left(\frac{x}{\sqrt{(1+t)}}\right) \quad \text{for } t \ge t_0.$$

(iii) $u_1(\cdot\,,t), u_2(\cdot\,,t)$ *are positive, classical solutions to* (1.1), (1.2) *with uniformly bounded* L^∞ *norms.*

Proof. (i) is an easy exercise, and we omit the proof. (ii) and (iii) are proved in Sec. 2.

Next, we observe that by making the change of variables:

$$w_i(y,s) = (1+t)u_i(x,t), \qquad i = 1,2, \quad s = \log(1+t), \quad y = \frac{x}{\sqrt{1+t}},$$

the system is changed to

$$\begin{aligned} w_{1,s} &= \triangle w_1 + \frac{y}{2}\cdot\nabla w_1 + w_1 - w_1 w_2, \\ w_{2,s} &= d\triangle w_2 + \frac{y}{2}\cdot\nabla w_2 + w_2 + w_1 w_2. \end{aligned} \tag{5.1}$$

It is a more convenient formulation to work with.

Lemma 5.2. *Suppose*

$$a_1 \in \mathcal{B}, \quad u_2(x,t) \ge c_0\phi(y)(1+t)^{-1},$$

then there exist $c_1 = c_1(c_0, q, d) > 0$ *and* $E = E(c_0, q, \|a_1\|)$ *such that*

$$u_1(x,t) \le E(1+t)^{-(2+c_1)/2}(1+|y|)^{-q}.$$

Proof. We use the formulation (5.1). The conclusion is the same as showing

$$w_1(y,s) \le e^{-sc_1/2}(1+|y|)^{-q}.$$

Let $D > 0$ be such that

$$D^2 > 4q(q+1)/(q-2)$$

and for all $|y| \geq D$,

$$\exp(-|y|^2/4) \leq |y|^{-q} \quad \text{and} \quad -|y|\exp(-|y|^2/4) \geq -2q|y|^{-q-1}.$$

Let $\overline{c} = \min_{|y|\leq D} c_0\phi(y)$. Without lost of generality, we assume $\overline{c} \leq (q-2)/4$. Denote

$$I(w) = w_s - \triangle w - \frac{y}{2}\cdot\nabla w - w + c_0\phi(y)w.$$

Let

$$\bar{w} = \begin{cases} E_1\exp(-|y|^2/4)e^{-c_1 s/2} + Me^{-c_1 s/2}, & |y| \leq D, \\ E_1\exp(-|y|^2/4)e^{-c_1 s/2} + Me^{-c_1 s/2}D^q|y|^{-q}, & |y| \geq D, \end{cases} \tag{5.2}$$

where $c_1 = \delta\overline{c}$ with $\delta > 0$ a small number such that $4c_1 < q-2$, and E_1 and M are positive numbers satisfying

$$c_1 E_1 < \frac{M}{4}(q-2-2c_1)e^{D^2/4}.$$

It is clear that with suitable choice of M, and consequently E_1, $\bar{w}(y,0) \geq w_1(y,0)$. We now demonstrate that $I(\bar{w}) \geq 0$, which, together with $I(w_1) \leq 0$, yields the conclusion of the lemma.

If $|y| \leq D$, it is easy to compute

$$I(\bar{w}) \geq e^{-c_1 s/2}(1-\delta)\overline{c}\big(E_1 e^{-|y|^2/4} + M\big) > 0.$$

If $|y| \geq D$, a more detailed calculation gives

$$\begin{aligned} I(\bar{w}) \geq e^{-c_1 s/2}\bigg(&\frac{-c_1}{2}E_1 e^{-|y|^2/4} + \frac{M}{2}(q-2-c_1)D^q|y|^{-q} \\ &-Mq(q+1)|y|^{-(q+2)}D^q\bigg) > 0 \end{aligned}$$

by our careful choice of c_1, D and E_1. This completes the proof of Lemma 5.2.

Lemma 5.3. *Suppose $a_1, a_2 \in \mathcal{B}$. There exists $\delta > 0$ such that*

$$\lim_{t\to\infty}\left|tu_2(x,t) - A\phi\left(\frac{x}{\sqrt{t}}\right)\right|t^\delta \to 0 \quad \textit{uniformly in } \{x : |x| \leq C\sqrt{t}\}, \tag{5.3}$$

$$\lim_{t\to\infty}\left|\int_{R^1} u_2(x,t)dx - A\right|t^\delta \to 0. \tag{5.4}$$

Proof. It is easy to see that $\|u_1 + u_2\|_1 = \|a_1 + a_2\|_1$ and $\|u_1\|_1$ is decreasing in t and $\|u_2\|_1$ is increasing. By Lemma 5.2,

$$\|u_1\|_1 \to 0 \quad \text{as } t \to \infty,$$

and therefore,

$$\|u_2\|_1 \to A \quad \text{as } t \to \infty.$$

Moreover, using the u_1 bound in Lemma 5.2, we deduce that u_2 satisfies, with some $\delta_0 > 0$,

$$u_{2,t} = \triangle u_2 + O(t^{-1-\delta_0})u_2.$$

Clearly, the nonlinear term is irrelevant and the conclusion follows from classical result for heat equation.

With the limiting profile of u_2 settled, we can now derive a better estimate for u_1 than the one given in Lemma 5.2.

Lemma 5.4. *Suppose $a_1, a_2 \in \mathcal{B}$ and $q > q(A)$. For any $\epsilon > 0$, there exists $t_0 > 0$ such that if $t \geq t_0$,*

$$u_1(x,t) \leq M(1+|y|)^{-q} t^{-\lambda-1},$$

where $\lambda = \lambda_{A-\epsilon}$ is the first eigenvalue of $\mathcal{L}_{\mathcal{A}-\epsilon}$ in $L^2(R^2, d\mu)$, and M is a positive constant.

Proof. Let $t_1 > 1$ be sufficiently large so that

$$w_2(y,s) \geq A\phi(y) - e^{-\delta s} \geq \frac{A}{2}\phi(y)$$

for $|y| \leq D$ and $s \geq s_1 \equiv \log t_1$, where D is a large positive number to be determined later. Set

$$\bar{w} = e^{-\lambda s}(M_1\psi(y) + (1+|y|^2)^{-q/2}), \tag{5.5}$$

where $\psi(y)$ is the eigenfunction of $\mathcal{L}_{\mathcal{A}-\epsilon}$ corresponding to λ with $\max_{y\in R^2}\psi(y) = 1$. Denote

$$J(w) = w_s - \Delta w - \frac{y}{2}\cdot\nabla w - w + w_2 w.$$

It is easy to see that

$$J(e^{-\lambda s}\psi(y)) \geq \begin{cases} e^{-\lambda s}\psi(y)[A\epsilon\phi(y) - e^{-\delta s}] & |y| \leq D, \\ -e^{-\lambda s}A\psi(y)\phi(y) & |y| \geq D. \end{cases} \tag{5.6}$$

Similarly,

$$\begin{aligned} &J(e^{-\lambda s}(1+|y|^2)^{-q/2}) \\ &\quad \geq e^{-\lambda s}\left(\left(\frac{q-2}{2} - \lambda\right)(1+|y|^2) - \frac{5}{2}q - \frac{q(q+2)|y|^2}{1+|y|^2}\right)(1+|y|^2)^{-q/2-1} \\ &\quad > e^{-\lambda s}(1+|y|^2)^{-q/2}\,(q-2-2\lambda)\,/4 > 0 \quad \text{if } |y| \geq D_1(\lambda, q), \end{aligned}$$

where D_1 is the first positive number such that

$$\frac{q-2-2\lambda}{4}(1+|y|^2)^2 \geq \frac{5}{2}q(1+|y|^2)+q(q+2)|y|^2 \quad \text{for all } |y| \geq D_1.$$

Here, we assume $D \gg D_1$. Clearly,

$$J(e^{-\lambda s}\psi(y)) \geq \frac{1}{2}e^{-\lambda s}A\epsilon\psi\phi > 0$$

if $s \geq s_2(A,D)$ and $|y| \leq D$. Furthermore,

$$J(M_1 e^{-\lambda s}\psi(y)) \geq \frac{1}{2}e^{-\lambda s}M_1 A\epsilon\psi\phi > 2q(q+2)e^{-\lambda s}(1+|y|^2)^{-q/2-1}$$

if $M_1 > M(A,q)$ and $|y| \leq D_1$. This shows that $J(\bar{w}) \geq 0$ if $|y| \leq D$ and $s \geq s_2(A,D)$.

Finally, if $|y| \geq D$ and D is sufficiently large,

$$\begin{aligned} &J\big(e^{-\lambda s}\big(M_1\psi(y)+(1+|y|^2)^{-q/2}\big)\big) \\ &\quad \geq \left(\left(\frac{q-2-2\lambda}{4}\right)(1+|y|^2)^{-q/2} - M_1 A\psi(y)\phi(y)\right)e^{-\lambda s} > 0. \end{aligned}$$

This shows that $\bar{w}$ is a super-solution if $s \geq s_0 \equiv \max(s_1, s_2)$. It is a simple matter to see that $w_1(y,s_0)$ can be bounded above by $M_2\bar{w}(y,s_0)$, where M_2 is a big positive number. By maximum principle $w_1(y,s) \leq M_2\bar{w}(y,s)$, $s > s_0$. This completes the proof of the lemma.

The success of the Renormalization Group method depends, crucially, on estimates of solutions to the linear operators $w_s = \mathcal{L}_{\mathcal{A}}w$ and $w_s = \mathcal{L}w$. We collect some relevant ones in the next lemma. In what follows, we use $e^{-s\mathcal{L}_{\mathcal{A}}}f$ and $e^{-s\mathcal{L}}f$ to denote their solutions with initial value f, respectively.

Lemma 5.5. *Suppose $f \in \mathcal{B}$.*

(i) *There exists $\delta = \delta(A,q)$ and $s_0 < \infty$ such that for $s \geq s_0$,*

$$\|e^{-s\mathcal{L}_{\mathcal{A}}}f\| \leq e^{-\delta s}\|f\|.$$

Moreover, if $q > q(A)$ and $(\psi_A, f) = 0$,

$$\|e^{-s\mathcal{L}_{\mathcal{A}}}f\| \leq e^{-(\delta+\lambda_A)s}\|f\|,$$

where λ_A is the first eigenvalue of $\mathcal{L}_{\mathcal{A}}$, and $(\cdot,\cdot)$ is the inner product in $L^2(R^2, d\mu)$.

(ii) *If $g \in L^2(R^2, d\mu)$, there exists $c > 0$ such that*

$$\|e^{-\mathcal{L}_{\mathcal{A}}}g\| \leq C\|g\|_2,$$

where $\|\cdot\|_2$ is the norm in $L^2(R^2, d\mu)$.

(iii) *Let* $\chi_u = \chi(|x| \geq \sigma)$, *the characteristic function. There exists* $C > 0$ *such that*

$$\|\chi_u e^{-\sigma \mathcal{L}_{\mathcal{A}}} g\| \leq C e^{-(q-1)\sigma/2} \|g\|.$$

(iv) *Suppose* $\int_{R^2} f dx = 0$. *Then, there exists* $\delta = \delta(q) > 0$ *and* $s_0 < \infty$ *such that for* $s \geq s_0$,

$$\|e^{-s\mathcal{L}} f\| \leq e^{-\delta s} \|f\|.$$

(v) *The quantities* $|\lambda_A - \lambda_{A'}|$, $|1 - (\psi_A, \psi_{A'})|$ *and the operator norm* $\|P_A - P_{A'}\|$ *in* $\mathcal{B}$ *are all bounded by* $C(M)|A - A'|$, *where* P_A *is the orthogonal projection in* $L^2(R^2, d\mu)$ *on* ψ_A *and* $\|P_A\| \leq C(M)$, *for* $0 \leq A,\ A' \leq M$.

Proof. The first part of (i) follows immediately from Lemma 5.2. The second part is proved in the proposition below.

To show (ii), observe that

$$(e^{-\mathcal{L}_{\mathcal{A}}} g)(x) = \int_{R^2} e^{-\mathcal{L}_{\mathcal{A}}}(x, y) e^{-y^2/8} e^{y^2/8} g(y)\, dy.$$

Then, an application of Cauchy–Schwartz inequality together with the fact that

$$\sup_x (1 + |x|)^q \left(\int_{R^2} \left(e^{-\mathcal{L}_{\mathcal{A}}}(x, y)\right)^2 e^{-y^2/4}\, dy \right) < \infty \tag{5.7}$$

yields the desired result. The validity of (5.7) can be verified using

$$e^{-s\mathcal{L}_{\mathcal{A}}}(x, y) \leq e^{-s\mathcal{L}_0}(x, y), \quad \text{by Feynmann–Kac formula}$$

and the Mehler's formula,

$$e^{-s\mathcal{L}_0}(x, y) = \left(4\pi(1 - e^{-s})\right)^{-1} \exp\left(-\frac{|x - e^{-s/2} y|^2}{4(1 - e^{-s})} \right).$$

Part (iii) is a direct consequence of a property of the "transformed" heat kernel $e^{-s\mathcal{L}_0}(x, y)$, see [7].

Part (iv) is essentially the same as the second part of (i) except the first eigenvalue is zero and the corresponding eigenfunction is $\phi = e^{-x^2/4d}$. The condition $\int_{R^2} f\, dx = 0$ means $(f, \phi) = 0$, where $(\cdot\,, \cdot)$ is the inner product in $L^2(R^2, \phi^{-1}\, dx)$.

Part (v) can be proved by using

$$(\psi_A, \psi_A') = 0, \quad \lambda_A' \leq (4\pi d)^{-1}$$

and $(\mathcal{L}_{\mathcal{A}} - \lambda_A)^{-1}$ is a bounded operator on the subspace $\{f \in \mathcal{B} \,|\, (f, \psi_A) = 0\}$, which follows from the second part of (i). For more details, see the appendix of [7]. Here, ψ_A' and λ_A' are derivatives of ψ_A and λ_A to A, respectively.

Remark 5.6. Some of the results in Lemma 5.5 is a direct extension to 2D of results in [7] for the 1D case, we collect them here for easy reference in the later proof.

Proposition 5.7. *The second part of* (i) *is true.*

Proof. Suppose $s_1 > 0$. Let $\delta > 0$ be small so that

$$q > 2 + 2\lambda_A + 4\delta \quad \text{and} \quad \mu_A - \lambda_A > 4\delta,$$

where μ_A is the second eigenvalue of $\mathcal{L}_{\mathcal{A}}$. Take $f \in \mathcal{B}$, $(\psi_A, f) = 0$ and $\|f\| = 1$. We proceed to show inductively that for $s_n = ns_1$, $\chi_b = \chi(|x| \le s_1)$ and $\chi_u = 1 - \chi_b$, $v(s_n) = e^{-s_n \mathcal{L}_{\mathcal{A}}} f$, we have the inequalities

$$\|\chi_u v(s_n)\| \le e^{-\beta n}, \tag{5.8}$$

$$\|\chi_b v(s_n)\|_2 + \|x_b v(s_n)\| \le e^{s_1^2/6} e^{-\beta n}, \tag{5.9}$$

where $\beta = (\lambda_A + 2\delta)s_1$. The conclusion of Proposition 5.1 follows immediately from (5.8) and (5.9) if n is large ($n\delta > s_1/6$) and $s = s_n$. For $s \in (s_n, s_{n+1})$, apply the first part of (i) in Lemma 5.5 and use the fact that $(n-1)\delta > s_1/6 + \lambda_A$ if n is large.

First, we note that if $f \in \mathcal{B}$ and $(\psi_A, f) = 0$, then

$$(e^{-s\mathcal{L}_{\mathcal{A}}} f, \psi_A) = 0 \quad \forall\, s > 0.$$

If $n = 0$, The bounds in (5.8) and (5.9) hold by the obvious inequality

$$\|\chi_b f\|_2 \le e^{s_1^2/8} \|f\| \tag{5.10}$$

and our assumption that $\|f\| = 1$.

If $n = 1$, (5.8) is true by (iii) of Lemma 5.5, since $q > 2 + 2\lambda_A + 4\delta$. Write

$$f = \chi_b f + \chi_u f = f_b + f_u.$$

Since $(\psi_A, f) = 0$,

$$\begin{aligned} |(f_b, \psi_A)| &= |(f_u, \psi_A)| \le C(A) s_1^{-(q-2-2\lambda_A)}, \\ \|e^{-s_1 \mathcal{L}_{\mathcal{A}}} f_b\|_2 &\le C(A) s_1^{-(q-2-2\lambda_A)} e^{-\lambda_A s_1} + e^{-\mu_A s_1} \|f_b\|_2 \le \frac{1}{4} e^{-\beta - \delta s_1} e^{s_1^2/8}, \end{aligned} \tag{5.11}$$

if s_1 is reasonably large. In the first inequality, we used the asymptotic behavior of ψ_A derived in Sec. 4. Using part (ii) of Lemma 5.5, we get

$$\|e^{-s_1 \mathcal{L}_{\mathcal{A}}} f_b\| \le C \|e^{-(s_1-1)\mathcal{L}_{\mathcal{A}}} f_b\|_2 \le C e^{-(s_1-1)(\lambda_A + 3\delta)} e^{s_1^2/8} < \frac{1}{4} e^{-\beta} e^{s_1^2/8}, \tag{5.12}$$

if s_1 is a reasonably large constant. By combining first part of (i) in Lemma 5.5 and (5.10), we have

$$\|\chi_b e^{-s_1 \mathcal{L}_{\mathcal{A}}} f_u\|_2 \le e^{s_1^2/8} \|e^{-s_1 \mathcal{L}_{\mathcal{A}}} f_u\| \le C e^{-\delta s_1} e^{s_1^2/8} \le \frac{1}{4} e^{-\beta} e^{s_1^2/6}. \tag{5.13}$$

It is clear that (5.11)–(5.13) gives (5.9).

Suppose (5.8) and (5.9) hold for n, we show they hold for $n+1$. Write $v = v(s_n) = v_b + v_u$. Since $(v, \psi_A) = 0$,

$$|(v_b, \psi_A)| = |(v_u, \psi_A)| \le C(A)\|v_u\| s^{-(q-2-2\lambda_A)} \le e^{-\beta n},$$

$$\|e^{-s_1\mathcal{L}_\mathcal{A}} v_b\|_2 \le e^{-\beta n} e^{-\lambda_A s_1} + e^{-\mu_A s_1}\|v_b\|_2 \le \frac{1}{4} e^{-(\beta+\delta s_1)(n+1)} e^{s_1^2/6}. \quad (5.14)$$

Again, by (ii) of Lemma 5.5, we get

$$\|e^{-s_1\mathcal{L}_\mathcal{A}} v_b\| \le C\|e^{-(s_1-1)\mathcal{L}_\mathcal{A}} f_b\|_2 \le Ce^{-(s_1-1)(\beta+\delta)(n+1)} e^{s_1^2/6} < \frac{1}{4} e^{-\beta(n+1)} e^{s_1^2/6}. \quad (5.15)$$

By combining first part of (i) in Lemma 5.5 and (5.10), we have

$$\|\chi_b e^{-s_1\mathcal{L}_\mathcal{A}} v_u\|_2 \le e^{s_1^2/8}\|e^{-s_1\mathcal{L}_\mathcal{A}} v_u\| \le Ce^{s_1^2/8} e^{-\delta s_1}\|v_u\| \le \frac{1}{4} e^{-\beta(n+1)} e^{s_1^2/6}. \quad (5.16)$$

It follows immediately that (5.14)–(5.16) implies (5.9) for $n+1$.

At last, (5.8) follows from

$$\|\chi_u e^{-s_1\mathcal{L}_\mathcal{A}} \chi_b g\| \le e^{-s_1^2/5}\|\chi_b g\|_2,$$

Lemma 5.5(iii) applied to $g = v_u$ and the bounds (5.8) and (5.9) for v_b and v_u. This completes the proof of proposition.

The RG method, as applied to our situation, is an iterative scheme. Let $L > 1$. We start by defining the RG map R:

$$(a_1, a_2) \to (a_1', a_2') \quad \text{in } \mathcal{B} \times \mathcal{B}$$

as

$$a_i'(x) = L^2 u_i(LX, L^2), \quad i = 1, 2,$$

where $a_i = u_i(x, 1)$, $i = 1, 2$ and (u_1, u_2) solve (1.1), (1.2). It is clear that

$$(a_1^n, a_2^n) \equiv R^n(a_1, a_2) = \big(L^{2n} u_1(L^n x, L^{2n}), L^{2n} u_2(L^n x, L^{2n})\big) = R(a_1^{n-1}, a_2^{n-1}).$$

Our ultimate goal is to prove that (u_1, u_2) behave asymptotically as the solution of the limiting linear problem. Accordingly, decompose the RG map as

$$a_1' = L^{-2\mathcal{L}_\mathcal{A}} a_1 + L^2 n_1(Lx, L^2), \quad (5.17)$$

$$a_2' = L^{-2\mathcal{L}} a_2 + L^2 n_2(Lx, L^2), \quad (5.18)$$

where

$$n_1(x,t) = -\int_1^t d\tau \int_{R^2} H_A(t,\tau,x,y)\left(u_1 u_2(y,\tau) - u_1 A\tau^{-1}\phi\left(\frac{y}{\sqrt{\tau}}\right)\right) dy,$$

$$n_2(x,t) = -\int_1^t d\tau \int_{R^2} H(t-\tau, x-y) u_1 u_2(y,\tau)\, dy,$$

where H_A is the fundamental solution of $u_t = \triangle u - A\phi u$ and A is to be defined in what follows.

To track the evolution, we write

$$a_1(x) = B\psi_A + b_1, \quad a_2(x) = A\phi(x) + b_2(x), \tag{5.19}$$

where $B = (a_1, \psi_A)$, $A = \int_{R^2} a_2\, dx$. With the normalization $(\psi_A, \psi_A) = 1$ and $\int_{R^2} \phi\, dx = 1$, we have

$$(\psi_A, b_1) = 0, \quad \int_{R^2} b_2\, dx = 0.$$

It is easy to verify that with same decomposition as in (5.19) for a_1', a_2',

$$A' = A + \int_{R^2} L^2 n_2(Lx, L^2)\, dx, \quad B' = (\psi_{A'}, a_1'),$$
$$b_1' = (1 - P_{A'})a_1', \quad b_2' = L^{-2\mathcal{L}} b_2 + L^2 n_2(Lx, L^2) + (A - A')\phi.$$

Since the system (1.1)–(1.2) is invariant under the scaling transform

$$u_{iK} = K u_i(Kx, K^2 t), \quad i = 1, 2$$

and Lemma 5.4, we can assume, for $1 \le s \le L^2$,

$$\|a_1\| < \epsilon, \quad \|a_1\| \cdot \|a_2\| < \epsilon, \quad \|u_1(s)\| < \epsilon, \quad \|u_1(s)\| \cdot \|u_2(s)\| < \epsilon.$$

We proceed to derive estimates for A', b_2', B' and b_1', in that order. Since,

$$\int_{R^2} H(t - \tau, x - y)(1 + |y|)^{-q}\, dy \le c e^{c(t-\tau)}(1 + |x|)^{-q}, \tag{5.20}$$

for some $c > 0$,

$$\|L^2 n_2(Lx, L^2)\| \le C(L)\epsilon \|a_2\| \quad \text{and} \quad \||n_2\|| = \sup_{t \in [1, L^2]} \|n_2(\cdot\,, t)\| \le C(L)\epsilon \|a_2\|. \tag{5.21}$$

It follows that

$$|A' - A| \le C(L)\epsilon \|a_2\|, \tag{5.22}$$

and consequently,

$$\|b_2'\| \le L^{-2\delta} \|b_2\| + C(L)\epsilon \|a_2\|$$

by (iv) of Lemma 5.5.

Now, consider n_1, and write

$$\begin{aligned} w(y, \tau) &= u_2(y, \tau) - A\tau^{-1}\phi\left(\frac{y}{\sqrt{\tau}}\right). \\ &= \left(e^{d(\tau-1)\Delta} b_2(y) + n_2(y, \tau)\right). \end{aligned}$$

By part (iv) of Lemma 5.5 and (5.21),

$$\||w\|| \le C(L)(\|a_2\| + A)(\|b_2\| + \epsilon\|a_2\|),$$

and, since $A \le C\|a_2\|$ and $\int_{R^2} H_A(t,\tau,x,y)(1+|y|)^{-q}\,dy \le C\Big(1+\frac{|x|}{\sqrt{t-\tau}}\Big)^{-q}$, which follows from Lemma 5.2, we obtain

$$\|Ln_1(Lx, L^2)\| \le C(L)\epsilon(\|b_2\| + \epsilon\|a_2\|), \tag{5.23}$$

using $\|u_1\|_\infty \le C\|a_1\|$ and $\|a_1\| \cdot \|a_2\| \le \epsilon$.

Next, we consider B'. Since from (5.17),

$$\begin{aligned} B' &= (\psi_{A'}, a_1') = (\psi_{A'}, L^{-2\mathcal{L}_A} a_1) + (\psi_{A'}, L^2 n_1(Lx, L^2)) \\ &= (\psi_{A'}, \psi_A) B L^{-2\lambda_A} + (\psi_{A'}, (P_{A'} - P_A) L^{-2\mathcal{L}_A} b_1) + (\psi_{A'}, L^2 n_1(Lx, L^2)), \end{aligned}$$

($P_A b_1 = 0$), we have, by (v) of Lemma 5.5 and (5.23),

$$\begin{aligned} |B' - BL^{-2\lambda_A}| &\le C|A' - A| L^{-2\lambda_A}(B + \|b_1\|) + C(L)\epsilon(\|b_2\| + \epsilon\|a_2\|) \\ &\le C(L)\epsilon^2 + C(L)\epsilon^2\|a_2\| + C(L)\epsilon\|b_2\|. \end{aligned}$$

In the last inequality, we used (5.22), $\|a_1\| < \epsilon$, $\|a_1\| \cdot \|a_2\| \le \epsilon$ and $|B| + \|b_1\| \le C\|a_1\|$. Finally, we estimate b_1'.

$$b_1' = (1 - P_{A'})a_1' = BL^{-2\lambda_A}(P_A - P_{A'})\psi_A + L^{-2\mathcal{L}_A} b_1 + L^{-2\mathcal{L}_A} b_1 (P_A - P_{A'}),$$

again by $P_A b_1 = 0$. It follows immediately from Lemma 5.5(i), (v), (5.23) and (5.22),

$$\|b_1'\| \le L^{-2(\lambda_A+\delta)}\|b_1\| + C(L)\epsilon(\epsilon(1 + \|a_2\|) + \|b_2\|).$$

To summarize our estimates, we have the following:

Lemma 5.8. *Suppose $L \ge L_0 = e^{2s_0}$, where s_0 satisfies the conditions in Lemma 5.4 and Lemma 5.5. There exists $\epsilon_0(L)$ and $C(L)$ such that if $\|a_1\| \cdot \|a_2\| \le \epsilon \le \epsilon_0(L)$, we have*

(a) $|A' - A| \le C(L)\epsilon\|a_2\|$,
(b) $\|b_2'\| \le L^{-2\delta}\|b_2\| + C(L)\epsilon\|a_2\|$,
(c) $|B' - L^{-2\lambda_A} B| \le C(L)\epsilon(\epsilon(1 + \|a_2\|) + \|b_2\|)$,
(d) $\|b_1'\| \le L^{-2(\lambda_A+\delta)}\|b_1\| + C(L)\epsilon(\epsilon(1 + \|a_2\|) + \|b_2\|)$.

Proof of Theorem 1.1. In light of results established in Secs. 2 and 3, we only need to show (i). We write a_i^n as in (5.19) with A_n, B_n and b_i^n in place of A, B and b_i, and derive bounds for A_n, B_n and b_i^n using Lemmas 5.4 and 5.8.

It is clear that

$$A_n \to A^* = \int_{R^2} (a_1 + a_2)\, dx.$$

First, by Lemma 5.4 and Lemma 5.8(a),

$$|A_{n+1} - A_n| \le C(L)\|a_1^n\| \cdot \|a_2^n\| \cdot \|a_2^n\| \le C(L)e^{-2n\eta},$$

with $0 < \eta < \lambda_A^*$, and hence,

$$|A_n - A^*| \le C(L)e^{-2n\eta}.$$

Set

$$n\lambda_n = \sum_{m=0}^{n-1} \lambda_{A_m}.$$

Since λ_A is a continuous and increasing function of A and A_n is a bounded and increasing sequence,

$$\lambda_{A_n} \to \lambda^* = \lambda_{A^*}, \quad \lambda_n \to \lambda_{A^*}.$$

If we take $\eta < \min(\delta, \lambda_{A^*})$, we get, by Lemma 5.8(b) and Lemma 5.4,

$$\|b_2^n\| \le e^{-2n\eta}\|a_2\| \quad \text{and} \quad \|b_1^n\| \le e^{-2n(\lambda_{A^*}+\eta)}(1 + \|a_2\|).$$

This, together with Lemma 5.8(c) and Proposition 2.4, gives

$$|B_{n+1} - B_n L^{-2\lambda_{A_n}}| \le C(L)L^{-2n(\lambda_{A^*}+\eta)}(1 + \|a_2\|),$$

with a smaller η, if necessary.

This shows there exists B^* such that

$$B_n L^{+2n\lambda_n} \to B^* \quad \text{as } n \to \infty.$$

Now, since λ_A is a differentiable function of A and $\lambda_A' \le 1/4\pi d$,

$$|n\lambda_n - n\lambda_{A^*}| = \left| \sum_{m=0}^{n-1} \lambda_{A_m} - \lambda_{A^*} \right| \le \sum_{m=0}^{n-1} |\lambda_{A_m} - \lambda_{A^*}| \le \sum_{m=0}^{n-1} C(L)e^{-2n\eta},$$

and therefore, $n\lambda_n - n\lambda_{A^*}$ converges to a finite limit as $n \to \infty$. Hence,

$$B_n L^{2n\lambda_{A^*}} \to B^{**} \quad \text{as } n \to \infty.$$

With $t = L^{2n}$, $L > L_0$, the results are directly translated to

$$\|tu_2(\sqrt{t}\,\cdot, t) - A^*\phi\| \le Ct^{-\eta}\|a_2\|, \tag{5.24}$$

$$\|t^{1+\lambda_{A^*}} u_1(\sqrt{t}\,\cdot, t) - B^{**}\psi_{A^*}\| \le Ct^{-\eta}(1 + \|a_2\|), \tag{5.25}$$

This proves Theorem 1.1.

Remark 5.9. To see why Theorem 1.1 cannot be true if $q \le q(A)$, we look at the linear problem

$$u_s = -\mathcal{L}_{\mathcal{A}} u. \tag{5.26}$$

Set

$$w_L(y,s) = \begin{cases} e^{-\lambda_A s}\psi_A(y) & |y| \le D, \\ Ce^{-\lambda_A s}(1+|y|^2)^{-q/2} & |y| \ge D, \end{cases}$$

where C is the constant which makes w_L continuous at $|y| = D$. It is easy to verify (see the proof of Lemma 5.3) that the function w_L is a sub-solution of (5.26) provided D is large enough such that

$$q(q+2)\frac{|y|^2}{(1+|y|^2)^2} \ge A^2\phi^2(y) \quad \forall\, |y| \ge D$$

and

$$\psi_A'(y) \le C\big((1+|y|^2)^{q/2}\big)' \quad \text{at} |y| = D.$$

It is clear that for a solution to (5.26) with continuous, positive initial value $u_0(y)$ satisfying

$$\lim_{|y|\to\infty} u_0(y)(1+|y|)^q = M > 0,$$

we can find a $\delta > 0$ such that $u_0(y) > \delta w_L(y,0)$. In consequence,

$$u(y,s) > \delta w_L(y,s), \quad s > 0.$$

This shows Theorem 1.1 cannot be true for the linear problem (5.26) since $w_L(y,s)$, though it has the desired time decay, has a much slower decay in space variable y than $\psi_A(y)$. In summary, in spite of the fact that we have not given a direct proof that Theorem 1.1 is false when $q \le q(A)$, the above serves as a strong evidence that it should not be true for the nonlinear problem if it is false for the limiting linear problem.

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